

Unit-4

Sampling and Statistical Inference

Population (Universe): The group of individual under study is called population. It may be finite or infinite.

Sampling: A finite subset of the population is called sample. The no. of individuals in the sample is called a sample size.

The process of selecting a sample from the population is called Sampling.

Random Sampling: The selection of individual from the population in such a way that each has the same chance of being selected.

Simple Sampling: It is a special case of Random Sampling in which trials are independent and probability of success is constant.

Sampling Distribution:

Consider, the samples of size 'n' drawn from the given population at random. The mean of these samples are not identical. If we group these means according to their frequencies, the frequency distribution so formed are called Sampling Distribution of means.

Symbols for Population and Samples:

Population

Size = N

Mean = μ

Standard

Deviation = σ

Proportion = P

Sample

Size = n

Mean = $\bar{x}$

Standard = s

Deviation

Proportion = $\bar{p}$

Standard Error: The standard deviation of the sampling distribution is called Standard Error.

The reciprocal of the Standard Error is called Precision.

NOTE:

- i) If $n \geq 30$ then the sample is said to be large, otherwise it is called small sample.
- ii) The sampling distribution of large samples is always assumed to be normal.

Types of Sampling:

1. Sampling with Replacement:

Here, the items are selected one by one and are put back to the population before the next draw.

Here, N^n possible samples can be drawn.

$\mu_{\bar{x}} \rightarrow$ the mean of the frequency distribution of the sample means.

$$\mu_{\bar{x}} = \frac{\sum f x}{\sum f}$$

$$\boxed{\mu_{\bar{x}} = \mu} \text{ and } \boxed{\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}} \Rightarrow \sigma_{\bar{x}} = \frac{\sum f(x - \bar{x})^2}{\sum f}$$

Here, $\sigma_{\bar{x}}$ is also known as standard error of mean.

2. Sampling without Replacement:

Here, the items are selected one by one and are not put back to the population before the next draw.

Here, $N_c n$ possible samples can be drawn.

$$\boxed{\mu_{\bar{x}} = \mu} \text{ and } \boxed{\sigma_{\bar{x}}^2 = \left(\frac{N-n}{N-1} \right) \frac{\sigma^2}{n}} \Rightarrow \sigma_{\bar{x}} = \underbrace{c}_{\text{correction factor}} \frac{\sigma}{n}$$

* When $n \gg$ large then 'c' is close to '1'.

1. The population consists of 4 numbers 3, 7, 11, 15. Consider all possible samples of size 2 which can be drawn: i) With replacement, ii) Without replacement from the population. Find: a) population mean, b) population standard deviation, c) The mean of standard deviation of means, d) The sampling error of means.

Given: $N=4$

$$\mu = \frac{3+7+11+15}{4} = 9$$

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{n} = \frac{(3-9)^2 + (7-9)^2 + (11-9)^2 + (15-9)^2}{4} = 20$$

$$\sigma = \sqrt{20} = 4.472$$

Sampling with replacement: $N=4$, sample size, $n=2$

Here, $4^2 = 16$ possible samples can be drawn.

$S = \{(3,7), (3,3), (3,11), (3,15), (7,3), (7,7), (7,11), (7,15), (11,3), (11,7), (11,11), (11,15), (15,3), (15,7), (15,11), (15,15)\}$

Mean of these samples:
 $x_i = \{3, 5, 7, 9, 11, 13, 15\}$
 $f_{x_i} = \{1, 2, 3, 4, 3, 2, 1\}$

$$\mu_{\bar{x}} = \frac{\sum (f_{x_i} x_i)}{\sum f_{x_i}} \Rightarrow \mu_{\bar{x}} = \frac{144}{16} = 9, \quad \sigma_{\bar{x}}^2 = \frac{\sum f_{x_i} (x_i - \bar{x})^2}{\sum f_{x_i}} = \frac{160}{16} = 10$$

For this distribution $\mu_{\bar{x}} = 9$ and $\sigma_{\bar{x}}^2 = 10 \Rightarrow \sigma_{\bar{x}} = 3.162 = \text{S.E.}$

$\therefore \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \Rightarrow \sigma^2 = n \sigma_{\bar{x}}^2$ Sampling Error of means
 $= 2(10) = 20$

Sampling without replacement:

$N C_n = 4 C_2 = 6$ possible samples can be drawn:

$S = \{(3,7), (3,11), (3,15), (7,11), (7,15), (11,15)\}$

Mean of these samples: $x_i = \{5, 7, 9, 11, 13\}$ frequency distribution table.
 $f_{x_i} = \{1, 1, 2, 1, 1\}$

$$\mu_{\bar{x}} = \frac{\sum (f_{x_i} x_i)}{\sum f_{x_i}} = \frac{54}{6} = 9, \quad \sigma_{\bar{x}}^2 = \frac{\sum f_{x_i} (x_i - \bar{x})^2}{\sum f_{x_i}} = \frac{40}{6} = \frac{20}{3}$$

For this distribution $\mu_{\bar{x}} = 9$ and $\sigma_{\bar{x}}^2 = \frac{20}{3} \Rightarrow \sigma_{\bar{x}} = 2.581$

$\therefore \sigma_{\bar{x}}^2 = \frac{\sigma^2 (N-n)}{n(N-1)} \Rightarrow \sigma^2 = n \sigma_{\bar{x}}^2 (N-1) / (N-n)$ S.E of means
 $= \frac{2(20)(4-1)}{3(4-2)} = \frac{120}{6} = 20$

Statistical Inference:

On the basis of sample information, we make certain decision about the population known as Statistical Inference.

Estimation and Hypothesis are the two aspects of Statistical Inference.

Estimation: To estimate the population parameters (μ, σ, σ^2 etc.) we use sample statistics such as $\bar{x}, s, s^2$ etc. Thus, Statistics is an estimation of parameter and this procedure is called Statistical Estimation.

Hypothesis: (Hypothesis Testing)
To reach decision about population on the basis of sample information we make certain decision about population. Such assumptions ^{assumptions} may or may not be true is called Hypothesis Testing.

Types of Hypothesis:

1. Null Hypothesis: It is based for the analysis of the problem: the hypothesis formulated for the purpose of rejecting it under the assumption of it is true is Null Hypothesis. It is denoted as ' H_0 '
2. Alternate Hypothesis: Any Hypothesis other than Null Hypothesis. It is denoted as ' H_1 '

One tailed and two tailed test:

The test of any statistical hypothesis where alternative hypothesis is one-sided i.e., if $H_0: \mu = \bar{x}$, then

$$H_1: \mu > \bar{x} \text{ (right tailed)}$$

$$H_1: \mu < \bar{x} \text{ (left tailed)}$$

is called one-tailed test.

A test of any statistical hypothesis where alternative hypothesis is two-sided i.e., if $H_0: \mu = \bar{x}$, then

$$H_1: \mu \neq \bar{x} \text{ is called two-tailed test.}$$

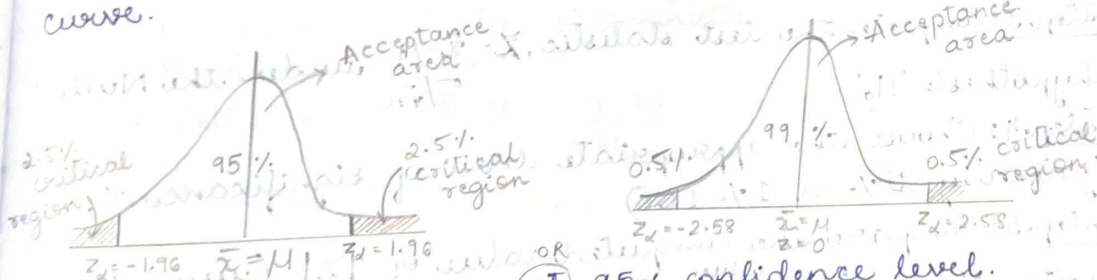
$$(H_1: \mu \neq \bar{x})$$

While testing the hypothesis, the following are the types of errors in decision taken.

Actually	Decision Taken	
	Accept H_0	Reject H_0
H_0 is True	Correct decision	(Wrong decision) Type 1 Error
H_0 is false	Type II Error	Correct Decision

Level of Significance:
The probability level below which led to rejection of the hypothesis is called Level of Significance.

The region in which a sample value, falling is rejected, is known as critical region. Generally we take two critical regions which cover 5% and 1% areas of the normal curve.



5% level of Significance at 95% confidence level.
1% level of Significance at 99% confidence level.

Test for large Samples ($n \geq 30$):
1. Z-test: Suppose that we've a normal population ' μ ' (mean) and ' σ ' (S.D). If ' $\bar{x}$ ' is the sample mean of a random sample of size ' n ' then $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ is called

standard normal variate. ($Z \rightarrow$ test statistic)

$\{ Z < Z_1 \rightarrow \text{Accept}$

$Z > Z_2 \rightarrow \text{Reject} \}$

From the normal distribution table, the confidence intervals for normal population are,

i) 95% confidence interval or 5% level of Significance

$$-1.96 < Z < 1.96$$

$$\text{i.e., } -1.96 < \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < 1.96$$

$$\Rightarrow \bar{x} - 1.96 \left[\frac{\sigma}{\sqrt{n}} \right] \leq \mu \leq \bar{x} + 1.96 \left[\frac{\sigma}{\sqrt{n}} \right]$$

ii) 99% confidence interval, or 1% level of significance

$$-2.58 < z < 2.58$$

$$\text{i.e., } -2.58 < \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} < 2.58$$

$$\Rightarrow \bar{x} - 2.58 \left[\frac{\sigma}{\sqrt{n}} \right] \leq \mu \leq \bar{x} + 2.58 \left[\frac{\sigma}{\sqrt{n}} \right]$$

Procedure for testing the statistical hypothesis:

Step 1: Setup the Null hypothesis ' H_0 '.

Step 2: Setup the Alternate hypothesis ' H_1 '. This will determine whether we've to use one-tailed or two-tailed test.

Step 3: Compute the test statistic, $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$ under the Null hypothesis ' H_0 '.

Step 4: Choose the appropriate level of significance α , (generally 5% or 1% LOS)

Step 5: Compare the computed value of $|Z|$ (using Step 3) with the critical value of z_α , at the required value of significance.

Step 6: If $|Z| < z_\alpha$, then accept the Null hypothesis H_0 . Otherwise $|Z| > z_\alpha$, then reject the Null hypothesis.

Central limit Theorem: $Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$

where, $\bar{x} \rightarrow$ mean of sample, $\mu \rightarrow$ mean of population

Note: If ' σ ' is not known then we use, $Z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$

where, s is Std. Deviation of sample.

1. A sample of 900 members is found to have a mean of 3.4 cm. Can it be reasonably regarded as a truly random sample from a large population with mean 3.25 cm and S.D. 1.61 cm.

→ Given; $n=900$, $\bar{x}=3.4\text{cm}$, $\mu=3.25\text{cm}$, $\sigma=1.61$.

Set, $H_0: \bar{x} = \mu$

By z-test, we've $z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$; $z = \frac{3.4 - 3.25}{1.61/\sqrt{900}} \Rightarrow z = 2.8 = z_{\text{cal}}$

under 5% level of significance, $z_{\alpha} = 1.96$ [$z_{\text{cal}} > z_{\alpha}$]

Reject H_0 .

under 1% level of significance, $z_{\alpha} = 2.58 = z_{0.01}$ [$z_{\text{cal}} > z_{0.01}$]

Reject H_0 .

In both the cases we can declare that the given sample is not the True Random Sample.

2. An ambulance service claims that it takes on the average less than 10 minutes to reach its destination in emergency calls. A sample of 36 calls has a mean of 11 minutes and the variance of 16 minutes. Test the significance at 0.05 level.

→ Given; $n=36$; $\bar{x}=11$ minutes, $\mu=10$ minutes, $s^2=16$ minutes
Set; $H_0: \bar{x} = \mu$ $S=4$ minutes

By z-test, we've $z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{11 - 10}{4/\sqrt{36}} \Rightarrow z = 1.5 = z_{\text{cal}}$

Under 5% loss: $z_{\alpha} = 1.96 = z_{0.05}$ ($z_{\text{cal}} < z_{0.05}$)
Accept H_0 .

Step 3

Test of Significance of Difference of Means:

Let μ_1 & μ_2 be the mean of 2 population ($\bar{x}_1, s_1$) & ($\bar{x}_2, s_2$) are the mean and standard deviation of 2 large samples of size n_1 & n_2 respectively.

To test the hypothesis for the difference of means we take ' H_0 ' as: $\mu_1 = \mu_2$ or $H_0: \bar{x}_1 = \bar{x}_2$

Here the Test statistic is $z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$

NOTE: If the samples are drawn from same population then

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

3. The means of two large samples of sizes 1000 and 2000 members are 67.5 inches and 68.0 inches respectively. Can the samples be regarded as drawn from the same population of S.D 2.5 inches?

→ Given; $\bar{x}_1 = \bar{x}_2$; $\mu_1 = 67.5$; $\sigma = 2.5$
 $n_1 = 1000, n_2 = 2000$ $\mu_2 = 68.0$

Set null hypothesis: $H_0: \bar{x}_1 = \bar{x}_2$ [no significant difference between mean of two samples]
 Also, $H_1: \bar{x}_1 \neq \bar{x}_2$

$$\text{By } z\text{-test: } z = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{67.5 - 68.0}{2.5 \sqrt{\frac{1}{1000} + \frac{1}{2000}}} = \frac{-0.5}{2.5 \sqrt{0.0015}} = \frac{-0.5}{2.5 \sqrt{0.0015}} = \frac{-0.5}{2.5 \times 0.0387} = \frac{-0.5}{0.09675} = -5.16$$

$$\Rightarrow z = -5.516 = z_{cal} \Rightarrow |z| = 5.516$$

Under 5% level of significance, $z_{\alpha} = z_{0.05} = 1.96 \Rightarrow |z| > z_{0.05}$
 Reject H_0

Under 1% level of significance, $z_{\alpha} = z_{0.01} = 2.58 \Rightarrow |z| > z_{0.01}$
 Reject H_0

4. Sample of students were drawn from two universities and from their weights in kilograms, mean and standard deviations are calculated and shown below. Make a large sample test to test the significance of the difference between the means.

	Mean	S.D	Sample size
University A	55	10	400
University B	57	15	100

→ Given; $n_1 = 400; n_2 = 100, \mu_1 = 55, \mu_2 = 57, s_1 = 10, s_2 = 15$

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{55 - 57}{\sqrt{\frac{100}{400} + \frac{225}{100}}} = \frac{-2}{\sqrt{0.25 + 2.25}} = \frac{-2}{\sqrt{2.5}} = \frac{-2}{1.581} = -1.26$$

$$|z| = 1.26$$

Under 5% level of significance, $z_{\alpha} = z_{0.05} = 1.96$
 $|z| < z_{0.05}$ Accept H_0

Under 1% level of significance, $z_{\alpha} = z_{0.01} = 2.58$
 $|z| < z_{0.01}$ Accept H_0

5. The mean yield of wheat from a district A was 210 pounds with S.D 10 pounds per acre from a sample of 100 plots. In another district the mean yield was 220 pounds with S.D 12 pounds from a sample of 150 plots. Assuming that the S.D of yield in the entire state was 11 pounds, test whether there is any significant difference between the mean yield of crops in the two districts.

→ Given; $\mu_1 = 210$, $S_1 = 10$ pounds/acre, $n_1 = 100$

$\mu_2 = 220$, $S_2 = 12$, $n_2 = 150$, $S = 11$

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{220 - 210}{\sqrt{\frac{100}{100} + \frac{144}{150}}} = 7.14$$

Under 5% LOS, $z_\alpha = z_{0.05} = 1.96 \Rightarrow |z| > z_{0.05}$ Reject H_0

Under 1% LOS, $z_\alpha = z_{0.01} = 2.58 \Rightarrow |z| > z_{0.01}$ Reject H_0

t-test: (for small samples)

Degree of freedom: It is the number of independent values which can be assigned to be statistical distribution.

Generally, degree of freedom $= \nu = n - k$

where, $n \rightarrow$ no. of observation

$k \rightarrow$ no. of constraints

Consider, a small sample of size 'n' drawn from a normal population with mean ' μ ' and S.D ' σ '. ' $\bar{x}$'s are the mean and S.D of the samples, then the test statistic is defined as

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$$

$$\text{where, } s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

This follows a student's 't' distribution with $(n-1)$ degree of freedom.

1. The nine items of a sample have the following values: 45, 47, 50, 52, 48, 47, 49, 53, 51. Does the mean of these differ significantly from the assumed mean of 47.5?

→ Given; $n = 9$, $\mu = 47.5$

$$\text{W.k.t; } \bar{x} = \frac{\sum x}{n} = \frac{442}{9} = 49.11$$

$$s^2 = \frac{\sum (x - \bar{x})^2}{n} = \left\{ \frac{\sum x^2}{n} - (\bar{x})^2 \right\} = \frac{21762}{9} - 2411.29$$

$$s^2 = 2418 - 2411.79 = 6.21 \Rightarrow s = 2.49$$

Set Null Hypothesis: $H_0: \mu = \bar{x}$

Set $H_1: \mu \neq \bar{x}$

By t-test, we've; $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{49.11 - 47.5}{2.49/\sqrt{8}} = \frac{1.61}{0.88} = 1.83$

$$t = 1.83 = t_{cal}$$

Here, dof (Degree of freedom) $\gamma = n - k = 9 - 1 = 8$

Under 5% level of significance, the table value for

$$t_{0.05} = 2.306 \text{ for } 8 \text{ dof}$$

$$\therefore |t_{cal}| < t_{0.05} \text{ Accept } H_0$$

Under 1% level of significance, the table value for

$$t_{0.01} = 3.08 \text{ for } 8 \text{ dof}$$

Hence, there's no significant difference between $\bar{x}$ & μ

2. A machinist is making engine parts with axle diameter of 0.7 inch. A random sample of 10 parts shows mean diameter 0.742 inch with a S.D of 0.04 inch. On the basis of the sample, would you say that the work is inferior?

Given; $n = 10$

$$\mu = 0.7$$

$$\bar{x} = 0.742$$

$$s = 0.04$$

By t-test we've; $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} = \frac{0.742 - 0.7}{0.04/\sqrt{9}} = \frac{0.042}{0.0133} = 3.15$

$$\Rightarrow t = 3.15 = t_{cal}$$

Here, dof (Degree of freedom) $\gamma = n - 1 = 10 - 1 = 9$

Under 5% level of significance, the table value for

$$t_{0.05} = 2.262 \text{ for } 9 \text{ dof}$$

$$\therefore |t_{cal}| > t_{0.05} \text{ Reject } H_0$$

Thus, Work is inferior.

Under 1% level of significance, the table value for

$$t_{0.01} = 3.250 \text{ for } 9 \text{ dof}$$

Hence, $t < t_{0.01} \Rightarrow$ Accept H_0

3. A certain stimulus administered to each of the 12 patients resulted in the following change in blood pressure: 5, 2, 8, -1, 3, 0, 6, -2, 1, 5, 0, 4. Can it be concluded that stimulus will increase the blood pressure?

→ Given; $n = 12$

Assume, $\mu = 0$ (for normal population)

$$\bar{x} = \frac{\sum x}{n} = \frac{2.583}{12}, \quad s^2 = \frac{\sum x^2}{n} - (\bar{x})^2 = \frac{185}{12} - 6.671$$

$$\Rightarrow s^2 = 15.416 - 6.671 = 8.745 \Rightarrow s = 2.95$$

By t-test we've; $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}} \Rightarrow t = \frac{2.583 - 0}{2.95/\sqrt{11}} = \frac{2.583}{0.889} = 2.90$

$$t = 2.9 = t_{cal}$$

Here, dof: $\gamma = 12 - 1 = 11$

Under 5% level of significance, the table value for

$$t_{0.05} = 2.201 \text{ for 11 dof}$$

$$\therefore |t_{cal}| > t_{0.05} \text{ Reject } H_0$$

Stimulus will increase the blood pressure.

Under 1% level of significance, the table value for

$$t_{0.01} = 3.106 \text{ for 11 dof}$$

$$\therefore t_{cal} < t_{0.01} \text{ Accept } H_0$$

Stimulus will not increase the blood pressure.

4. Eleven school boys were given a test in work shop drawing. They were given a month's further tuition and a second test of equal difficulty was held at the end of it. Do the marks give evidence that the students have benefitted by extra coaching?

Boys	1	2	3	4	5	6	7	8	9	10	11
Marks I test	23	20	19	21	18	20	18	17	23	16	19
Marks II test	24	19	22	18	20	22	20	20	23	20	17

→ Given; $n = 11$, $\mu = 0$ (Assume)

Difference in marks of I and II test;

-1, 1, -3, 3, -2, -2, -2, -3, 0, -4, 2

Let $H_0: \bar{x} = \mu$ (no significant difference b/w marks)

Also $H_1: \bar{x} \neq \mu$

$$\Rightarrow \bar{x} = \frac{\sum x}{n} = \frac{-11}{11} = -1$$

$$\Rightarrow s^2 = \frac{\sum x^2}{n} - (\bar{x})^2 = \frac{61}{11} - 1$$

$$\Rightarrow s^2 = 4.545 \text{ (50/11)}$$

By t-test: $t = \frac{\bar{x} - \mu}{s/\sqrt{n-1}}$

$$\Rightarrow t = \frac{-1 - 0}{\sqrt{4.545}} = -1.485 \Rightarrow |t| = 1.485$$

$$\text{O.P.B.} = \frac{2.5137}{\sqrt{100}} = 0.0678$$

Here, $\text{dof} = \gamma = n - 1 = 10$

Under 5% level of significance, the table value for

$$t_{0.05} = 2.23 \text{ for } 10 \text{ dof}$$

$$\therefore |t| < t_{0.05} \Rightarrow \text{Accept } H_0$$

(Thus, the students are not benefitted by extra coaching)

• 95% confidence limits for population mean μ

$$|t| \leq t_{0.05}$$

$$\left| \frac{\bar{x} - \mu}{s/\sqrt{n-1}} \right| \leq t_{0.05}$$

$$\bar{x} - \frac{s}{\sqrt{n-1}} (t_{0.05}) \leq \mu \leq \bar{x} + \frac{s}{\sqrt{n-1}} (t_{0.05})$$

• The heights of 10 males of a given locality are found to be 175, 168, 155, 170, 152, 170, 175, 160, 160, 165 cms. Based on this sample, find the 95% confidence limits for the height of males in that locality.

→ Given; $n = 10$

$$\bar{x} = \frac{\sum x}{n} = \frac{1650}{10} = 165$$

$$s^2 = \frac{\sum x^2}{n} - (\bar{x})^2 \Rightarrow s^2 = \frac{272828}{10} - 272250 = 57.8$$

$$\Rightarrow s = 7.6$$

W.K.T: Confidence limits (95%) for mean ' μ ';

$$\bar{x} - \frac{s}{\sqrt{n-1}} (t_{0.05}) \leq \mu \leq \bar{x} + \frac{s}{\sqrt{n-1}} (t_{0.05})$$

But, $\gamma = n-1 = 10-1 = 9$ d.o.f, $t_{0.05} = 2.26$

$$\Rightarrow 165 - \frac{7.6}{3} (2.26) \leq \mu \leq 165 + \frac{7.6}{3} (2.26)$$

$$\Rightarrow 159.27 \leq \mu \leq 170.72 \approx \boxed{159 \leq \mu \leq 171} \text{ is the confidence limit}$$

Test for the difference b/w means of two samples:

If $\bar{x}_1$ and $\bar{x}_2$ are the means of two independent samples ' n_1 ' and ' n_2 ' respectively from a normal population with mean ' μ ' and S.D ' σ ', then we set Null Hypothesis that $H_0: \bar{x}_1 = \bar{x}_2$ (no significant difference between means of two samples)

Alternate Hypothesis: $H_1: \bar{x}_1 \neq \bar{x}_2$

Then by t-test,
$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

If ' σ ' is unknown then we estimate it from samples.

If s_1 and s_2 are the S.D's of the samples then ' σ '

can be found using
$$\sigma = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$$

Thus,
$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}} \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

It follows Student's t distribution with $\boxed{\gamma = n_1 + n_2 - 2}$ degree of freedom.

Suppose if the samples sizes are equal ($n_1 = n_2 = n$) then,

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{n(s_1^2 + s_2^2)}{2n-2}} \left(\frac{2}{n}\right)} \Rightarrow \boxed{t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2 + s_2^2}{n-1}}}$$

2. Samples of two types of electric bulbs were tested for length of life and the following data were obtained.

	Size	Mean	SD
Sample 1	8	1234h	36h
Sample 2	7	1036h	40h

Is the difference in the means sufficient to warrant that type 1 bulbs are superior to type 2 bulbs?

→ Given; $n_1 = 8$, $n_2 = 7$, $s_1 = 36$

and $\bar{x}_1 = 1234$, $\bar{x}_2 = 1036$, $s_2 = 40$

Set $H_0: \bar{x}_1 = \bar{x}_2$ also $H_1: \bar{x}_1 \neq \bar{x}_2$

(no difference in lifespan of two samples)

By t-test, we've: $t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$

$$\text{Consider; } \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} = \frac{8(36)^2 + 7(40)^2}{8 + 7 - 2} = \frac{10368 + 11200}{13}$$

$$= 1659.076$$

$$\frac{1}{n_1} + \frac{1}{n_2} = \frac{1}{8} + \frac{1}{7} = 0.2678$$

$$\text{Hence, } t = \frac{1234 - 1036}{\sqrt{1659.076 \times 0.2678}} = \frac{198}{40.731 \times 0.5174} = 9.4$$

$$\text{dof, } n_1 + n_2 - 2 = 7 + 8 - 2 = 13$$

Under 5% level of significance, the table value

$$t_{0.05} = 2.160 \text{ for } 13 \text{ dof}$$

Here, $t > t_{0.05}$ Reject H_0

Hence, Type 1 bulbs are superior to Type 2 bulbs.

3. Two independent samples of sizes 8 and 7 contained the following Table

Sample 1: 19, 17, 15, 21, 16, 18, 16, 14

Sample 2: 15, 14, 15, 19, 15, 18, 16

Is the difference between the sample means significant

→ Given: $n_1 = 8$, $n_2 = 7$

Set $H_0: \bar{x}_1 = \bar{x}_2$ (no significant difference between the means of 2 samples)

$$H_1: \bar{x}_1 \neq \bar{x}_2$$

By t-test; we've $t = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ $\Rightarrow \sigma = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$

from the given table: $\bar{x}_1 = \frac{\sum x_1}{n_1} = \frac{136}{8}$, $\bar{x}_2 = \frac{\sum x_2}{n_2} = \frac{112}{7}$

$$\bar{x}_1 = 17, \bar{x}_2 = 16$$

$$\Rightarrow s_1^2 = \frac{\sum x_1^2 - (\bar{x}_1)^2}{n_1} = \frac{2348 - (289)}{8} = 4.5 \Rightarrow s_1 = \sqrt{4.5} = 2.12$$

$$\Rightarrow s_2^2 = \frac{\sum x_2^2 - (\bar{x}_2)^2}{n_2} = \frac{1812 - (256)}{7} = 2.857 \Rightarrow s_2 = \sqrt{2.857} = 1.69$$

$$\therefore \sigma = \sqrt{\frac{8(4.5) + 7(2.857)}{8+7-2}} = \sqrt{4.307} = 2.075$$

$$\therefore t = \frac{17-16}{2.075 \sqrt{\frac{1}{8} + \frac{1}{7}}} = \frac{1}{2.075(0.496)} = 0.971$$

$$\text{dof, } n_1 + n_2 - 2 = 7 + 8 - 2 = 13$$

Under 5% level of significance, the table value

$$t_{0.05} = 2.160 \quad \text{for } 13 \text{ dof}$$

Here, $t < t_{0.05}$ Accept H_0

Hence, there's no significant difference b/w mean of two samples.

4. The mean height and the SD height of 8 randomly chosen soldiers are 166.9 and 8.29 cm respectively. The corresponding values of 6 randomly chosen sailors are 170.3 and 8.50 cm respectively. Based on this data, can we conclude that the soldiers are, in general, shorter than sailors?

→ Given; $n_1 = 8$ and $n_2 = 6$

$$\bar{x}_1 = 166.9, \bar{x}_2 = 170.3$$

$$s_1 = 8.29, s_2 = 8.50$$

Let $H_0: \bar{x}_1 = \bar{x}_2$ (no significant difference)

$$H_1: \bar{x}_1 \neq \bar{x}_2$$

By t-test; $t = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$ $\Rightarrow \sigma = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$

$$\Rightarrow \sigma = \sqrt{\frac{988.29}{12}} = \sqrt{81.94} = \underline{9.05}$$

$$\therefore t = \frac{166.9 - 170.3}{9.05 \sqrt{0.2916}} = \frac{-3.4}{4.887} = \underline{-0.695}$$

Under 5% level of significance, the table value for $t_{0.05} = 2.17$ for 12 dof

Here, $t < t_{0.05}$ Accept H_0

Hence in general the soldiers are shorter than sailors.

5. The table values gives biological values for 6 cow's milk & 6 buffalo's milk. Examine whether the differences are significant.

Cow's Milk 1.8 2.0 1.9 1.6 1.8 1.5

Buffalo's Milk 2.0 1.8 1.8 2.0 2.1 1.9

→ By t-test; $t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2 + s_2^2}{n-1}}}$

Here, $n=6$

$$\Rightarrow \bar{x}_1 = \frac{\sum x_1}{n_1} = \frac{10.6}{6} = \underline{1.76}$$

$$\Rightarrow \bar{x}_2 = \frac{\sum x_2}{n_2} = \frac{11.6}{6} = \underline{1.93}$$

$$\Rightarrow s_1^2 = \frac{\sum x_1^2}{n_1} - (\bar{x}_1)^2 = \frac{18.9}{6} - 3.097$$

$$\Rightarrow s_2^2 = \frac{\sum x_2^2}{n_2} - (\bar{x}_2)^2 = \frac{22.5}{6} - 3.7249$$

$$s_1^2 = \underline{0.0521}$$

$$s_2^2 = \underline{0.025}$$

$$\sigma = \sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}}$$

$$\Rightarrow \sigma = \sqrt{\frac{6(s_1^2 + s_2^2)}{6+6-2}}$$

$$t = \frac{1.76 - 1.93}{\sqrt{0.0154}} = \frac{-0.17}{0.124} = \underline{-1.37}$$

Under 5% level of significance, the table value for $t_{0.05} = 2.228$ for 10 dof

Here, $t < t_{0.05}$ Accept H_0

Thus, there's no significant difference b/w samples.

Test of significance of variance of two samples:

F-test: Let; $x_1, x_2, \dots, x_{n_1}$ and $y_1, y_2, \dots, y_{n_2}$ be the values of two independent normal samples drawn from normal population having equal variance ' σ^2 ' then we define F-statistic by the relation,

$$F = \frac{\frac{n_1 s_1^2}{n_1 - 1}}{\frac{n_2 s_2^2}{n_2 - 1}} \quad \text{where } s_1^2 = \frac{\sum (x - \bar{x})^2}{n_1}$$

$$s_2^2 = \frac{\sum (y - \bar{y})^2}{n_2}$$

This gives F-distribution (variance-ratio distribution) with $(n_1 - 1, n_2 - 1)$ degree of freedom. Here we set Null Hypothesis as $H_0: s_1^2 = s_2^2$

NOTE:

The larger of variance must be placed in numerator.

Test the equality of standard deviations for the data given below at 5% level of significance $n_1 = 10, n_2 = 14, s_1 = 1.5, s_2 = 1.2$.

→ Here; $s_1^2 > s_2^2$

Set $H_0: s_1^2 = s_2^2$ (no significant difference between variance of two samples)

$$H_1: s_1^2 \neq s_2^2$$

By F-test: $F = \frac{n_1 s_1^2}{n_1 - 1} = \frac{22.5}{9} \Rightarrow F = 1.612$

$$\frac{n_2 s_2^2}{n_2 - 1} = \frac{20.16}{13}$$

Under 5% level of significance, the value for (9, 13) dof is $F_{0.05} = 2.71$

$\therefore F < F_{0.05}$ Accept H_0

Two sample sizes 8 and 7 give the sum of squares of deviations from their respective means equal to 34 and 24 respectively. Test the hypothesis that the populations have the same variance given that $F_{0.05} = 4.2$ for (7, 6) dof.

→ Given; $n_1 = 8, n_2 = 7$

$$\sum (x - \bar{x})^2 = 34, \sum (y - \bar{y})^2 = 24$$

$$\text{W.k.t; } s_1^2 = \frac{\sum (x_i - \bar{x})^2}{n_1} = \frac{34}{8} = 4.25$$

$$s_2^2 = \frac{\sum (y_i - \bar{y})^2}{n_2} = \frac{24}{7} = 3.42$$

Here; $s_1^2 > s_2^2$

Set $H_0: s_1^2 = s_2^2$ (no significant difference)

$H_1: s_1^2 \neq s_2^2$

$$\text{By F-test; } F = \frac{n_1 s_1^2}{n_1 - 1} = \frac{8(4.25)}{7} = \frac{4.857}{3.99} = 1.217$$

$$\frac{n_2 s_2^2}{n_2 - 1} = \frac{7(3.42)}{6}$$

Under 5% level of significance, the table value for (7, 6) dof in F-test is $F_{0.05} = 4.2$

$\therefore F_{0.05} > F$ Accept H_0

Hence, there's is no significant difference between variance of two samples.

3. The nicotine contents in two random samples of tobacco are given below.

Sample 1 21 24 25 26 27

Sample 2 22 27 28 30 36 31

Can you say that the two samples came from the same population?

→ Given; $n_1 = 5$; $n_2 = 6$; $\bar{x}_1 = 123/5 = 24.6$

$$\text{W.k.t; } s_1^2 = \frac{\sum (x_i - \bar{x}_1)^2}{n_1} \Rightarrow s_1^2 = \frac{(-0.6)^2 + (-3.6)^2 + (0.4)^2 + (1.4)^2 + (2.4)^2}{5}$$

$$\Rightarrow s_1^2 = \frac{21.2}{5} = 4.24$$

$$\bar{x}_2 = 174/6 = 29; s_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2} = \frac{(-7)^2 + (-2)^2 + 1 + 1 + 49 + 4}{6}$$

$$s_2^2 = \frac{108}{6} = 18$$

Here, $s_1^2 < s_2^2$

Set $H_0: s_1^2 = s_2^2$ (no significant difference)

$H_1: s_1^2 \neq s_2^2$

$$\text{By F-Test; } F = \frac{n_2 s_2^2}{n_2 - 1} \Rightarrow F = \frac{6(18)}{5} = \frac{21.6}{5.3}$$

$$\frac{n_1 s_1^2}{n_1 - 1} = \frac{5(4.24)}{4}$$

$$\Rightarrow F = 4.07$$

Under 5% level of significance, the table value for (4,5) dof in F-test is $F_{0.05} = 5.26$ $\therefore F_{0.05} > F$ Accept H_0
 Yes, the two samples came from the same population.

Two independent samples of sizes 7 and 6 have the following values:

Sample A 28 30 32 33 33 29 34

Sample B 29 30 30 24 27 29

Examine whether the samples have been drawn from normal population having the same variance?

Given; $n_1 = 7$, $n_2 = 6$

$$\bar{x}_1 = \frac{\sum x_1}{n_1} = \frac{219}{7} = 31.28$$

$$s_1^2 = \frac{\sum (x_1)^2}{n_1} - (\bar{x}_1)^2 = \frac{6883}{7} - 978.43$$

$$s_1^2 = 4.85$$

$$\bar{x}_2 = \frac{\sum x_2}{n_2} = \frac{169}{6} = 28.16$$

$$s_2^2 = \frac{\sum (x_2)^2}{n_2} - (\bar{x}_2)^2 = \frac{4787}{6} - 792.98$$

$$s_2^2 = 4.84$$

Here, $s_1^2 > s_2^2$

Let $H_0: s_1^2 = s_2^2$ (no significant difference)

$H_1: s_1^2 \neq s_2^2$

$$\text{By F-test: } F = \frac{n_1 s_1^2}{n_1 - 1} \div \frac{n_2 s_2^2}{n_2 - 1} \Rightarrow F = \frac{7(4.85)}{6} \div \frac{6(4.84)}{5} \Rightarrow F = \frac{5.233}{5.808} = 0.9$$

Under 5% level of significance, the table value for (6,5) dof in F-test is $F_{0.05} = 4.95$ $\therefore F_{0.05} > F$ Accept H_0

Thus, the samples drawn from normal population have same variance.

Two random samples drawn from 2 normal populations are given below. Test whether the 2 populations have the same variance.

Sample I 20 16 26 27 23 22 18 24 25 19

Sample II 17 23 32 25 22 24 28 6 31 33 20 27

→ Given; $n_1 = 10$, $n_2 = 12$

$$\bar{x}_1 = \frac{\sum x_1}{n_1} = \frac{220}{10} = 22 \quad s_1^2 = \frac{\sum x_1^2}{n_1} - (\bar{x}_1)^2 = \frac{4960}{10} - 484$$

$$\bar{x}_2 = \frac{\sum x_2}{n_2} = \frac{288}{12} = 24 \quad s_2^2 = \frac{\sum x_2^2}{n_2} - (\bar{x}_2)^2 = \frac{7526}{12} - 576$$

$$s_2^2 = 57.16$$

Here, $s_2^2 > s_1^2$

Set $H_0: s_1^2 = s_2^2$ (no significant difference)

$$H_1: s_1^2 \neq s_2^2$$

By F-Test: $F = \frac{n_1 s_1^2}{n_1 - 1} \Rightarrow F = \frac{57.16(12)}{11} = 62.35$
 $F = \frac{n_2 s_2^2}{n_2 - 1} = \frac{12(10)}{9} = 13.33$
 $F = 4.677$

Under 5% level of significance, the table value for (11, 9) dof in F-Test is $F_{0.05} = 3.1$ $\therefore F > F_{0.05}$ Reject H_0

Under 1% level of significance, the table value for (11, 9) dof in F-Test is $F_{0.01} = 5.19$ $\therefore F_{0.01} > F$ Accept H_0

χ^2 -test: (Test for goodness of fit)

It can be used to test the goodness of fit. This test can also be applied to test the compatibility of observed and theoretical frequencies (expected frequencies)

Let; $O_1, O_2, \dots, O_n$ be the observed frequencies and $e_1, e_2, \dots, e_n$ be the corresponding expected frequencies such that $\boxed{\sum O_i = \sum e_i = N}$ where $N = \text{no. of members of population}$.

The test statistics is; $\boxed{\chi^2 = \sum_{i=1}^n \frac{(O_i - e_i)^2}{e_i}}$ $n \rightarrow \text{no. of terms}$
 $c \rightarrow \text{no. of constraints}$

with $(n - c)$ degree of freedom.

Here we set Null hypothesis $H_0: \sum O_i = \sum e_i$ that is, Theoretical and observed frequencies are compatible.

Conditions for the validity of χ^2 test:

- i) The observations should be independent (N).
- ii) The total frequency $N \geq 50$ (must be reasonably large).
- iii) The theoretical frequency $e_i \geq 5$ (individual frequency).
In case if $e_i < 5$ it should be pooled with the adjacent frequency, so that the combined frequency must be ≥ 5 .
- iv) If any parameter is estimated from the observed distribution then the value of 'c' would be 1 more than the no. of parameters.

The theory predicts that the proportion of beans available in 4 groups A, B, C, D should be 9:3:3:1. In an experiment with 1600 beans the numbers in the 4 groups were 882, 313, 287 and 118. Use χ^2 -test to verify whether the experimental result supports the theory.

→ Given: $N = 1600$, $n = 4$ { 882, 313, 287, 118 }
 O_1 O_2 O_3 O_4

Expected frequencies:

$$e_1 = \frac{9}{16} \times 1600 = 900; e_2 = \frac{3}{16} \times 1600 = 300; e_3 = \frac{3}{16} \times 1600 = 300$$

$$e_4 = \frac{1}{16} \times 1600 = 100$$

no significant difference between
Set $H_0: O_i = e_i$ (observed frequencies & expected frequencies)

$$H_1: O_i \neq e_i$$

$$\text{By } \chi^2\text{-test: } \chi^2 = \sum_{i=1}^n \frac{(O_i - e_i)^2}{e_i}$$

$$\Rightarrow \chi^2 = \sum_{i=1}^4 \frac{(O_i - e_i)^2}{e_i} = \frac{(882 - 900)^2}{900} + \frac{(313 - 300)^2}{300} + \frac{(287 - 300)^2}{300} + \frac{(118 - 100)^2}{100}$$

$$\Rightarrow \chi^2 = 0.36 + 0.563 + 0.563 + 3.24 = 4.726 \approx 4.73$$

Under 5% level of significance, the table value for

$$\chi_{0.05}^2 = 7.85 \quad \text{with } \gamma = n - 1 = 4 - 1 = 3$$

$$\therefore \chi^2 < \chi_{0.05}^2 \quad \text{Accept } H_0$$

Thus, observed and expected frequencies are compatible.

2. Among 64 offspring of a certain cross between Guinea pigs 34 were red, 10 were black and 20 were white. According to the genetic model these numbers should be in the ratio 9:3:4. Are the data consistent with the model at 5% level?

→ Given: $N = 64$, $n = 3$; $O_1 = 34$, $O_2 = 10$, $O_3 = 20$

Expected frequencies:

$$e_1 = \frac{9}{16} \times 64 = \underline{36}, \quad e_2 = \frac{3}{16} \times 64 = \underline{12}, \quad e_3 = \frac{4}{16} \times 64 = \underline{16}$$

Set $H_0: O_i = e_i$ (no significant difference)

$H_1: O_i \neq e_i$

By χ^2 -test: $\chi^2 = \sum_{i=1}^3 \frac{(O_i - e_i)^2}{e_i}$

$$\Rightarrow \chi^2 = 0.11 + 0.33 + 1 = \underline{1.44}$$

Under 5% level of significance, the Table value for

$$\chi^2_{0.05} = 5.99 \quad \text{for 2 dof}$$

$$\therefore \chi^2_{0.05} > \chi^2_{0.05} \quad \text{Accept } H_0$$

Hence, the data is consistent.

3. Five Coins are Tossed, 320 Times. The number of heads observed is given below, whether the coin is unbiased.

No. of heads 0 1 2 3 4 5 Total

Frequency 15 45 85 95 60 20 320

→ Given: $N = 320$, $n = 5$; $O_1 = 15$, $O_2 = 45$, $O_3 = 85$, $O_4 = 95$,
 $O_5 = 60$, $O_6 = 20$

Let us apply binomial distribution to find the expected frequencies, i.e., $P(x) = {}^n C_x p^x q^{n-x}$ $\{p = \frac{1}{2}, q = \frac{1}{2}, n = 5\}$

When 5 coins are tossed, the expected frequencies 0, 1, 2, 3, 4, 5... in 320 trials are

$$\Rightarrow e_1 = 320 \times {}^5 C_0 \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^5 = \underline{10}$$

$$\Rightarrow e_2 = 320 \times {}^5 C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^4 = \underline{50}$$

$$\Rightarrow e_3 = 320 \times {}^5 C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^3 = \underline{100}$$

$$\Rightarrow e_4 = 320 \times {}^5 C_3 \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \underline{100}$$

$$\Rightarrow e_5 = 320 \times {}^5 C_4 \left(\frac{1}{2}\right)^4 \left(\frac{1}{2}\right) = \underline{50}$$

$$\Rightarrow e_6 = 320 \times {}^5 C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right)^0 = \underline{10}$$

Set $H_0: O_i = e_i$ (no significance difference)

$H_1: O_i \neq e_i$

By χ^2 -Test: $\chi^2 = \sum_{i=1}^5 \frac{(O_i - e_i)^2}{e_i}$

$\Rightarrow \chi^2 = 2.5 + 0.5 + 2.25 + 0.25 + 2 + 10 = 17.5$

Under 5% level of significance, the table value for

$\chi^2_{0.05} = 9.49$ for 4 dof

$\therefore \chi^2_{0.05} < \chi^2$ Reject H_0

thus, the observed and theoretical frequencies are not compatible under 5% level of significance.

4. Fit a poisson distribution for the following data and test the goodness of fit.

$x:$	0	1	2	3	4	5	6	Total
$f:$	273	170	30	7	7	2	1	390

(d) b

(e) P

(b) 88

(c) Fl

$OP = 88 + Fl + b + P = 6 + 0 + 0 + 0 = 11$

$1 - b = (1 - 0)(1 - 0) = (1 - 0)(1 - 0) = 1$

$88 - b, 0 - Fl, 0 - P, 0 - 0$

$\chi^2 = \frac{\sum (O_i - e_i)^2}{e_i}$

$\frac{OP + Fl + b + P}{(b + 0)(0 + 0)(0 + 0)(0 + 0)}$

$OP, Fl = 0$

Under 5% level of significance, the table value for

$\chi^2_{0.05} = 8.8$

$\chi^2 > \chi^2_{0.05} \therefore$ Reject H_0

therefore, no fit is observed.

$\chi^2_{0.05} = 10.0$

$\chi^2 < \chi^2_{0.05} \therefore$ Accept H_0

• χ^2 -Test for the independent attributes

It is applicable when it is subjected to the following conditions;

- i) The observation should be independent.
- ii) The total frequency 'N' should be reasonably large.
- iii) Each of the frequencies are independent frequencies.

(a, b, c, d) ≥ 5 then,

$$\text{Test statistic, } \chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}$$

with $\gamma = 1$ dof $\{ \gamma = (n-1)(m-1) \}$

1. The following table shows the results of an experiment to investigate the effect of vaccination induced on the animals against a particular disease. Use χ^2 -test to test the hypothesis that there is no difference between vaccinated and unvaccinated groups. (i.e., groups and disease are independent)

	Got disease	Did not get disease
vaccinated	9 (a)	42 (b)
Not vaccinated	17 (c)	28 (d)

→ Set, H_0 : Vaccination and disease are independent.

$$N = a + b + c + d = 9 + 42 + 17 + 28 = 96$$

$$\gamma = (n-1)(m-1) = (2-1)(2-1) = 1 \text{ dof}$$

Here; a = 9, b = 42, c = 17, d = 28

$$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)} = \frac{96(213444)}{4176900}$$

$$\chi^2 = 4.90$$

Under 5% level of significance, the table value for

$$\chi^2_{0.05} = 3.84$$

with $\gamma = 1$ dof

$$\therefore \chi^2_{0.05} < \chi^2 \Rightarrow \text{Reject } H_0 \text{ (check for 1\%)}$$

Hence, vaccination has no effect on animals.

$$[\chi^2_{0.01} = 6.63 \text{ with } \gamma = 1 \text{ dof}]$$

$$\therefore \chi^2_{0.01} > \chi^2 \Rightarrow \text{accept } H_0$$

2. In a survey result

→ Let, H_0

H_1

$$N = a +$$

$$\gamma = (n -$$

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→ Set, H_0

H_1

$N =$

In order to test whether smoking and literacy are independent a survey of 210 literates and 250 illiterates was conducted. The result of the survey is as follows.

	Smokers	Non-Smokers
Literates	13	197
Illiterates	46	204

Let, H_0 : Smoking and literacy are independent.

H_1 : Smoking and literacy are not independent.

$$N = a + b + c + d = 13 + 46 + 197 + 204 = 460$$

$$\gamma = (n-1)(m-1) = (2-1)(2-1) = 1 \text{ dof}$$

$$\text{Here, } a = 13, b = 197, c = 46, d = 204.$$

$$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)} \Rightarrow \chi^2 = \frac{460(41088100)}{1242097500}$$

$$\chi^2 = 15.216$$

Under 5% level of significance, the table value for

$$\chi^2_{0.05} = 3.84 \text{ with } \gamma = 1 \text{ dof}$$

$$\therefore \chi^2_{0.05} < \chi^2, \text{ Reject } H_0$$

Hence, Smoking and literacy are ^{not} independent.

2. In a school there were 180 students in class 10th. Among them 128 students passed in preparatory examination, where as 153 students passed in the final board exam. However, those who passed the preparatory exam only 122 students passed in the final examination. Test at 1% level of significance, whether passing in the preparatory exam and final exam are independent.

		Preparatory		
		Pass	Fail	
Final	Pass	122	31	→ 153
	Fail	6	21	→ 27
		↓	↓	
		128	52	180 Total

Let, H_0 : Passing in preparatory & final examinations are independent.

H_1 : ——— not independent.

$$N = a + b + c + d = 180$$

$$\gamma = (n-1)(m-1) = 1 \text{ dof}$$

$$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)} = \frac{180(5645316)}{27495936}$$

$$\chi^2 = 36.95$$

ii) Under 1% level of significance, the table value for

$$\chi^2_{0.01} = 6.63 \text{ with } \gamma = 1 \text{ dof}$$

iii) $\chi^2_{0.01} < \chi^2$ Reject H_0

Hence, Passing in preparatory & final exam are not independent.

$$\chi^2 = \frac{(0.018801H)0.02H}{0.037602H \cdot 1} = \frac{(0.0004 - 0.0001)H}{(0.1H)(0.1H)(0.1H)(0.1H)} = \frac{0.0003H}{0.0001H} = 3H$$

for 1% level of significance, the table value for 1 dof is 3.84

$$\chi^2_{0.01} = 3.84 \text{ with } 1 \text{ dof}$$

$\chi^2 > \chi^2_{0.01} \therefore$ Reject H_0

Interpretation: There is a significant difference between the number of students who passed in the first exam and the number of students who passed in the preparatory exam. The number of students who passed in the first exam is significantly higher than the number of students who passed in the preparatory exam.

	Preparatory	Final
Pass	15	16
Fail	16	0
Total	31	16

Interpretation: There is a significant difference between the number of students who passed in the preparatory exam and the number of students who passed in the final exam. The number of students who passed in the preparatory exam is significantly higher than the number of students who passed in the final exam.